Thomas Nguyễn

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PROFESSIONAL SUMMARY

- AI Engineer specializing in Computer Vision, Deep Learning, and MLOps pipelines.
- Strong proficiency in Python and PyTorch, combined with practical experience in automation testing framework engineering and advanced Generative AI applications (RAG & Multi-Agent systems)

TECHNICAL SKILLS

- **AI & Generative AI:** PyTorch, OpenCV, LlamaIndex, CrewAI, Mediapipe, Scikit-learn, LangGraph
- **Automation & MLOps:** Ansible (YAML), Jenkins, DVC, MLflow, Airflow, Docker, Docker Compose, GitHub Actions.
- **Backend & Cloud:** Python (FastAPI, Flask), Celery, Redis, PostgreSQL, Azure, Linux.
- **Monitoring & Tools:** Prometheus, Grafana, Jira, Bitbucket, Git.

EDUCATION

Ton Duc Thang university (TDTU) Ho Chi Minh City, Viet Nam
Bachelor of Computer Science - GPA: 7.3/10 *September 2020 - September 2025*

Coursework: Object-Oriented Programming, Data Structure and Algorithms, Database Management Systems, Machine Learning, Computer Vision, Natural Language Processing

Honors:

- Second prize for students researching integration with Computer Vision at Ton Duc Thang University (2021 - 2022)
- Second Prize in the competition on correcting IPA pronunciation for Vietnamese learners (2024–2025)

WORK EXPERIENCES

Associate Software Engineer April 2025 - Present
Terralogic Company *Ho Chi Minh City, Viet Nam*

Core Infrastructure & Automation:

- Designed, optimized, and maintained complex **Ansible playbooks (YAML)** to automate system configuration and functional validation.
- Integrated automated Regression Test suites into **Jenkins CI/CD pipelines**, leveraging **Bitbucket** and **Jira** to accelerate delivery cycles and sustain high infrastructure stability.

AI R&D Initiative (Speech-to-IPA Product):

- Awarded Second Prize in the company's internal AI development competition for independently pioneering an end-to-end Speech-to-IPA pronunciation correction system.
- Fine-tuned a **deep learning** model using **PyTorch**, integrating it with a **Flask** backend and **Next.js** frontend to build the functional product.
- Optimized local inference to achieve low-latency execution with high precision, rigorously evaluating model performance using **Word Error Rate (WER)** and **Character Error Rate (CER)** metrics.

AI Engineer Intern August 2024 - November 2024
TMA Solution Lab 6 *Ho Chi Minh City, Viet Nam*

- Developed a sign language recognition model, achieving **94% accuracy** using **CNN1D** + **Transformer**.
- Preprocessed data using **Mediapipe** & **OpenCV**, extracting 543 landmark points and applying **Z-score normalization**.
- Optimized model performance through **K-fold cross-validation**, **Focal Loss**, and **Affine Data Augmentation.**
- Built **Flask APIs** and converted **PyTorch** models to **ONNX** format for optimized inference.
- Evaluated model effectiveness using the **Confusion Matrix**, **F1-score**, **Precision**, and **Recall**.
- Collaborated in an **Agile** team, utilizing **Git** for source code version control.

PROJECTS

Health Chatbot for Childcare Support

- Built an advanced RAG chatbot using **LlamaIndex**, **Groq API**, and **ChromaDB** (Vector Database) to optimize document embedding storage and semantic search retrieval.
- Processed unstructured data (PDFs/Web) via **BeautifulSoup** and deployed a production-ready interface using **Streamlit** and **FastAPI**.
- **Source**: Code

Personalized Morning Research Agent

- Architected a **LangGraph** state-machine agent integrated with a **PostgreSQL** database to manage user profile states and session persistence, automating end-to-end research pipelines.
- Integrated **Groq API** for rapid inference alongside **DuckDuckGo API** for real-time aggregation; built an automated multi-format email delivery pipeline via Gmail SMTP, containerized with **Docker Compose**.
- **Source**: Code

CERTIFICATIONS

IELTS - IDP September 2024 - September 2026